

Object Oriented Programming

BSIT - Fall 2024

Semester 3

Quiz - 8

Time Allowed: Mins 20

Total Marks: 20

Scenario: Implement **Name** and **Date** helper classes and a **Member** main class that composes them.

What to implement (only the class bodies — *no main()*):

1. Name helper class

- **Private:** `char* fullName;`
- **Public:** default ctor, `Name(const char* s)` (deep copy), copy ctor, destructor printing `"~Name()"`, `void setName(const char*)`, `const char* getName() const`.

2. Date helper class

- **Private:** `int day, month, year;`
- **Public:** default ctor, `Date(int d=1,int m=1,int y=2000)`, copy ctor, destructor printing `"~Date()"`, `void setDate(int d,int m,int y)`, `void getDate(int& d,int& m,int& y) const` (or a display).

3. Member main class

- **Private:** `Name name; Date dob; int id;`
- **Public:**
 - Parameterized constructor: `Member(const char* nameStr, int d, int m, int y, int idVal)` — do not use member-init list; set helper objects via their setters inside body.
 - Copy constructor (deep-copy through helpers, no member-init list).
 - Destructor printing `"~Member()"`.
 - `void setName(const char*)`, `void setDOB(int d,int m,int y)`, `void display() const`.

Constraints & grading hints:

- **Name** must manage dynamic memory correctly (deep copy, proper delete).
 - **Member** must use composition (helpers as objects).
 - No member-init lists.
 - Constructors/destructors should print messages to demonstrate order.
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